OSI Models

عناصر أي عملية اتصال (Communication Elements)
أي اتصال شبكي لازم يحتوي:

- **Source (Sender)** المرسل
- **Destination (Receiver)** المستقبل
- **Channel (Medium)** الوسيط أو المسار (Cable, Wireless, Fiber)

[EXAM TIP] وجود اتصال فيزيائي فقط غير كافي، لازم اتفاق على "كيف" يتم الاتصال

Communication Protocols

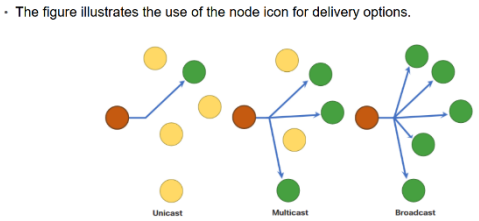
- **All communications are governed by protocols**
- البروتوكولات = **قواعد الاتصال**
- تختلف القواعد حسب البروتوكول المستخدم

Message Delivery Options
طرق إرسال الرسائل:

Unicast	One-to-One
Multicast	One-to-Many (not all)
Broadcast	One-to-All

IMPORTANT:
Broadcast is used in IPv4
- IPv6 does NOT support Broadcast
+ IPv6 uses Anycast

رمز العقدة (Node) غالبًا **دائرة**
يستخدم لتمثيل أي جهاز (PC, Router, Server)



Network Protocol Overview
Can be implemented on devices in :
البروتوكولات يمكن تنفيذها :

- Software
- Hardware
- Both

كل بروتوكول له :

- Function
- Format
- Rules

Types of Network Protocols

Protocol Type	Purpose
Network Communication	Enable device-to-device communication
Network Security	Authentication, Integrity, Encryption
Routing	Exchange routing info & choose best path
Service Discovery	Automatic detection of devices/services

Network Protocol Functions

Function	Description
Addressing	Identify sender & receiver
Reliability	Guaranteed delivery
Flow Control	Efficient data rate
Sequencing	Order segments

Function	Description
Error Detection	Detect corrupted data
Application Interface	Process-to-process communication

[VERY IMPORTANT] TCP provides Reliability, Sequencing, Flow Control

Protocol Interaction (Stacking)

Protocol	Function
HTTP	Client–Server interaction & content format
TCP	Manage conversations & reliability
IP	Global delivery
Ethernet	Local LAN delivery

Common Protocol Suites

- **TCP/IP** – Most common (IETF)
- **OSI** – ISO & ITU (Reference model)
- AppleTalk – Proprietary (Apple)
- Novell NetWare – Proprietary (Novell)

TCP/IP Protocol Suite

- Used by the **Internet**
- **Open standard**
- Vendor-independent
- Ensures **interoperability**

[EXAM TIP] TCP/IP is NOT owned by a company

TCP/IP Communication Process

- Server **encapsulates**
- Client **de-encapsulates**

Open Standards Encourage

- Interoperability
- Competition
- Innovation

Characteristics

- Vendor-neutral
- Non-profit
- Promote open standards

Internet Standards Organizations

Organization	Role
ISOC	Promotes internet development
IAB	Manages internet standards
IETF	Develops TCP/IP
IRTF	Long-term research

ICANN & IANA

- **ICANN**: IP addresses, domain names
- **IANA**: Manages IPs, domains, protocol numbers

Other Standards Bodies

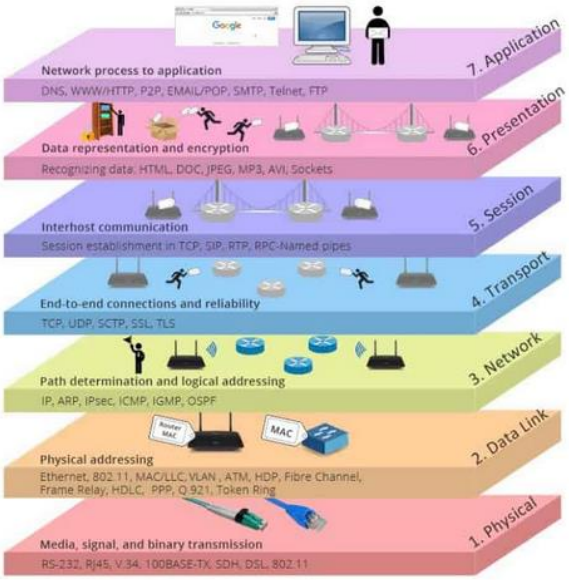
IEEE – Networking, telecom •
EIA – Wiring & racks •
TIA – VoIP, radio, cellular •
ITU-T – IPTV, DSL, video compression •

Why Layered Models?

Simplify complex networks •
Standard language •
Vendor interoperability •
Changes in one layer don't affect others •

OSI Model (7 Layers)

Layer	Name	Function
7	Application	User services
6	Presentation	Data format
5	Session	Session management
4	Transport	Segmentation & delivery
3	Network	Logical addressing
2	Data Link	Framing & MAC
1	Physical	Bits & signals



TCP/IP Model

Layer	Function
Application	User data
Transport	End-to-end
Internet	Routing
Network Access	Media & hardware

TCP/IP does NOT define physical media protocols

Segmenting

Break data into smaller units
Benefits :

Faster transmission •
Efficient retransmission •

Sequencing

Numbering segments •
TCP is responsible •

Encapsulation is the process where protocols add their information to the data.
At each stage of the process, a PDU has a different name to reflect its new functions.
There is no universal naming convention for PDUs, in this course, the PDUs are named according to the protocols of the TCP/IP suite.
PDUs passing down the stack are as follows :

Data (Data Stream) •
Segment •
Packet •
Frame •
Bits (Bit Stream) •

Encapsulation & De-encapsulation

- **Encapsulation**: Top → Down
- **De-encapsulation**: Bottom → Up

Addressing Responsibilities

- **Layer 3 (IP)**: End-to-end delivery
- **Layer 2 (MAC)**: Local delivery (hop by hop)

Same Network

- Same network portion
- Destination MAC = actual device MAC

Remote Network

- Destination MAC = **Default Gateway**
- IP stays the same
- MAC changes every hop

IMPORTANT: IP address = global (unchanged) MAC address = local (changes per hop)
Both the data link and network layers use addressing to deliver data from source to destination.
Network layer source and destination addresses: Responsible for delivering the IP packet from original source to the final destination.
Data link layer source and destination addresses : Responsible for delivering the data link frame from one network interface card (NIC) to another NIC on the same network.

Physical Layer L1

Physical Connection
قبل أي اتصال شبكي لازم يوجد **اتصال فيزيائي**
الاتصال قد يكون:

- Wired
- Wireless

Network Interface Card (NIC) يتم عبر
بعض الأجهزة تحتوي:

- واحدNIC
- أو أكثر (Wired + Wireless)

[EXAM POINT] Not all physical connections provide the same performance
What the Physical Layer Does

- **Transports bits** across network media
- يستقبل **Frame كامل** من Data Link Layer
- يقوم بـ **Encoding** وتحويله إلى Signals

الجهاز التالي:

- يستقبل bits
- يعيد encapsulation
- يقرر ماذا يفعل بها

IMPORTANT: Physical Layer works with bits, not frames or packets
Three Functional Areas :

1 **Physical Components**

- Hardware
- NICs
- Interfaces

- Connectors
- Cable materials & design

2 Encoding

- Converts bits stream to recognizable format
- Provides predictable patterns
- Encoding methods:
  - Manchester
  - 4B/5B
  - 8B/10B

[EXAM FAVORITE] Encoding ≠ Signaling

3 Signaling

- How **1** and **0** are represented
- Depends on media type:
  - Electrical (Copper)
  - Light pulses (Fiber)
  - Microwave (Wireless)

Copper Cabling

Characteristics

- Most common
- Cheap
- Easy to install

Limitations

- **Attenuation** (signal weakens)
- **EMI / RFI**
- **Crosstalk**

Mitigation

- Cable length limits
- Shielding & grounding
- Twisted pairs

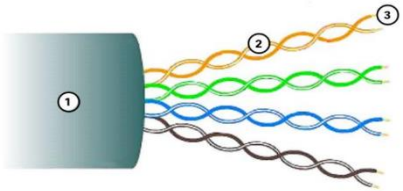
Types of Copper Cabling

- **UTP**
- **STP**
- **Coaxial**

UTP (Unshielded Twisted Pair)

- Most common
- RJ-45 connectors
- Used between hosts & devices

Key Characteristics

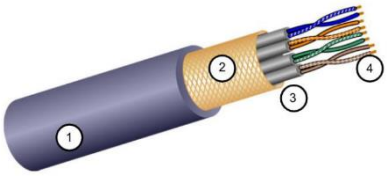


١. Outer jacket protection
٢. Twisted pairs reduce interference
٣. Color-coded insulation

STP (Shielded Twisted Pair)

- Better noise protection
- More expensive
- Uses shielding (foil / braid)

Key Characteristics

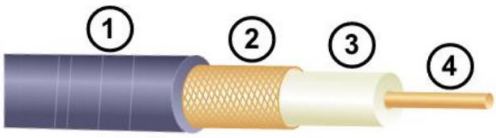


١. Outer jacket protection
٢. Braided or foil
٣. Foil shield for each pair
٤. Color-coded insulation

Coaxial Cable

Outer cable jacket to prevent minor physical damage -2

2- A woven copper braid, or metallic foil, acts as the second wire in the circuit and as a shield for the inner conductor.
3- A layer of flexible plastic insulation
4- A copper conductor is used to transmit electronic signals



Used for the following :
Wireless antennas ○
Cable Internet ○

UTP Cabling

How UTP Reduces Crosstalk

1 **Cancellation**

Opposite polarity •
Magnetic fields cancel •

2 **Different twist rates**

Each pair twisted differently •

UTP Standards

TIA/EIA-568 •
Cable type ○
Length ○
Termination ○
IEEE rates performance •

Categories

Cat 3 •
Cat 5 / 5e •
Cat 6 •

Connectors

RJ-45 Connector •
RJ-45 Socket •

Cable Types

Cable	Usage
Straight-Through	Host → Switch
Crossover	Switch → Switch (Legacy)
Rollover	Console access

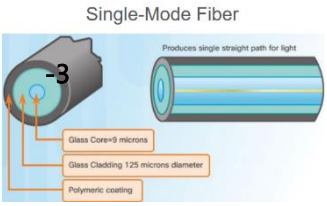
Fiber-Optic Cabling

Properties

High bandwidth •
Long distance •
Immune to EMI/RFI •
Uses **light** •
Glass core •
Laser or LED •

Types of Fiber

Single-Mode Fiber (SMF)

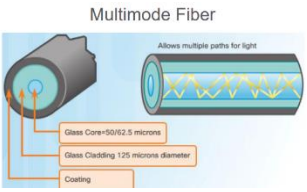


-3

Small core •
Expensive laser •
Long distance •

2- Multimode Fiber (MMF)

Larger core •



- LEDs
- Up to **10 Gbps / 550 m**
- More dispersion

SMF = longer distance . MMF = shorter distance .

Fiber Usage

- Enterprise backbone
- FTTH
- Long-haul networks
- Submarine cables

Fiber Connectors

- ST
- SC
- LC (Simplex / Duplex)

Fiber Patch Cords

- Yellow = Single-Mode
- Orange / Aqua = Multimode

Fiber vs Copper

Feature	UTP	Fiber
Bandwidth	Low–Medium	Very High
Distance	Short	Very Long
EMI/RFI	Affected	Immune
Cost	Cheap	Expensive

Wireless Media

Properties

- Radio / Microwave
- Highest mobility

Limitations

- Coverage issues
- Interference
- Security risks
- Shared medium
- Half-duplex

Wireless Standards

Standard	Use
IEEE 802.11	Wi-Fi
IEEE 802.15	Bluetooth
IEEE 802.16	WiMAX
IEEE 802.15.4	Zigbee (IoT)

Wireless LAN (WLAN)

Requires:

- Wireless Access Point
- Wireless NIC

[VERY IMPORTANT] WLAN security policies are mandatory

Routers and computers understand BINARY, humans use DECIMAL.

IPv4 = 32 bits Octet = 8 bits

Hexadecimal is used for Ipv6 and MAC addresses.

Data Link Layer L2

طبقة **Data Link (Layer 2)** مسؤولة عن :

- **Hop-to-Hop delivery**
- التواصل بين **NIC ↔ NIC**
- تحويل **Layer 3 Packets** إلى **Layer 2 Frames**
- استخدام **Physical (MAC) Addressing**
- ورفض الإطارات التالفة **Error Detection**

IMPORTANT: Layer 2 = Local delivery only (NOT end-to-end)

Access to Physical Media

- تُمكّن الطبقات العليا من استخدام الـPhysical Layer
- مسؤولة عن **Encapsulation / De-encapsulation** للإطارات (Frames)

عند كل Hop :

1 يستقبل Frame

2 يفك التغليف(De-encapsulation)

3 يعيد التغليف(Re-encapsulation)

4 يرسل Frame جديد للـNext Hop

[EXAM FAVORITE] MAC address changes at every hop

Two Sublayers

1 **Logical Link Control (LLC)**

يربط بين :

- Software (Upper Layers)
- Hardware (Lower Layers)

2 **Media Access Control (MAC)**

- Data encapsulation
- Media access control
- Access methods:
  - CSMA/CD
  - CSMA/CA

[VERY IMPORTANT] LLC = Software communication MAC = Hardware & media access

IEEE 802 LAN/MAN Standards

تختلف حسب نوع الشبكة :

- Ethernet (802.3)
- WLAN (802.11)
- WPAN (802.15)

Data Link Layer Standards Bodies

تعريف بروتوكولات Layer 2 يتم بواسطة :

- **IEEE**
- **ITU**
- **ISO**
- **ANSI**

Physical vs Logical Topology

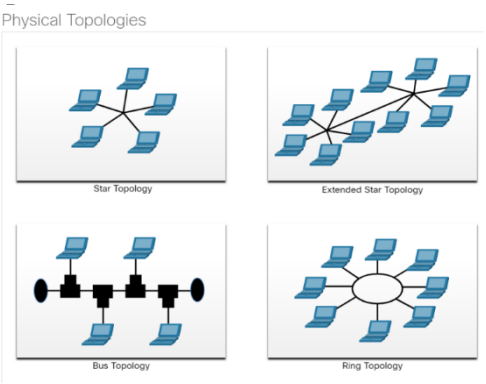
Type	Description
Physical	التوصيل الفعلي للأسلاك
Logical	كيف تنتقل البيانات فعليًا

WAN Topologies

Topology	Description
Point-to-Point	اتصال مباشر بين جهازين
Hub-and-Spoke	موقع مركزي مع فروع
Mesh	أعلى توفر – أعلى تكلفة

[EXAM POINT] Point-to-Point = simplest WAN topology

LAN Topologies



- **Star**
- **Extended Star**

(Legacy) Old :

- Bus
- Ring

Modern Ethernet uses Star topology

- Bus and Ring are legacy

Duplex Communication

Half-Duplex -4

إرسال أو استقبال فقط

مستخدم في :

- WLAN
- القديمة Ethernet hubs

2- Full-Duplex

إرسال واستقبال بنفس الوقت

مستخدم في Ethernet switches

[EXAM FAVORITE] Switches operate in Full-Duplex

All nodes operating in half-duplex, competing for use of the medium. Examples are :

Carrier sense multiple access with collision detection (CSMA/CD) as used on legacy bus-topology Ethernet.

Carrier sense multiple access with collision avoidance (CSMA/CA) as used on Wireless LANs.

CSMA/CD (Collision Detection)

يستخدم في Ethernet القديم

عند التصادم :

1 اكتشاف التصادم

2 انتظار وقت عشوائي

3 إعادة الإرسال

CSMA/CA (Collision Avoidance)

- مستخدم فيWireless
- يتجنب التصادم قبل حدوثه
- الأجهزة تُعلن مدة الإرسال

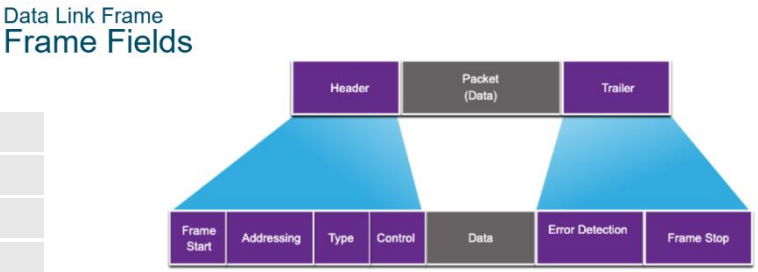
IMPORTANT: Ethernet detects collisions and Wireless avoids collisions

Data Link Frame :

Frame Structure

يتكون من Frame :

- Header
- Data
- Trailer



Frame Fields

Field	Function
Frame Start/Stop	بداية ونهاية
Addressing	Source & Destination
Type	Layer 3 protocol
Control	Flow control
Data	Payload
Error Detection	كشف الأخطاء

[EXAM FAVORITE] Error detection is done at Layer 2

Layer 2 Addresses (MAC)

MAC Address خصائص

- **Physical Address** يُسمى
- Frame Header موجود في
- يُستخدم فقط داخل الشبكة المحلية
- **يتغير عند كل جهاز يمر عليه**

Reminder: MAC = Local || IP = End-to-End

The logical topology and physical media determine the data link protocol used :

- Ethernet
- Wireless 802.11
- Point-to-Point (PPP)
- High-Level Data Link Control (HDLC)
- Frame-Relay

[EXAM POINT] The data link protocol depends on topology & media

---

Ethernet Switching L2

Ethernet Encapsulation

: يعمل في Ethernet

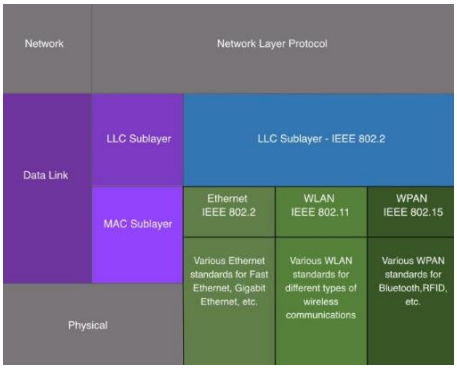
- **Data Link Layer**
- **Physical Layer**

: معرف بواسطة معايير

- **IEEE 802.2** (LLC)
- **IEEE 802.3** (Ethernet)

[EXAM POINT] Ethernet is NOT a single protocol, it is a family of technologies

: يستخدم طبقتين فرعيتين Ethernet



1 LLC – Logical Link Control (802.2)

يحدد: أي **Layer 3 protocol** مستخدم داخل الإطار

: يعمل كـ Interface بين

- Network Layer
- MAC sublayer

2 MAC – Media Access Control

:مسؤول عن

- Data Encapsulation
- Media Access
- MAC Addressing

MAC = framing + addressing + media access , LLC = identifies Layer 3 protocol

MAC Sublayer Responsibilities

- Ethernet Frame structure
- Source & Destination MAC
- **FCS** باستخدام Error detection

Media Access (CSMA/CD)

: القديم Ethernet

- Bus topology
- Hubs
- **Half-Duplex**
- **CSMA/CD** يستخدم

: الحديث Ethernet

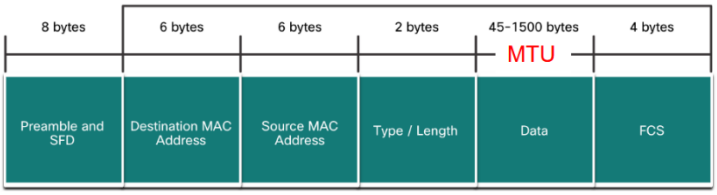
- Switches
- **Full-Duplex**
- لا يستخدم CSMA/CD

[EXAM FAVORITE] CSMA/CD is NOT used in switched full-duplex Ethernet

Ethernet Frame Fields :

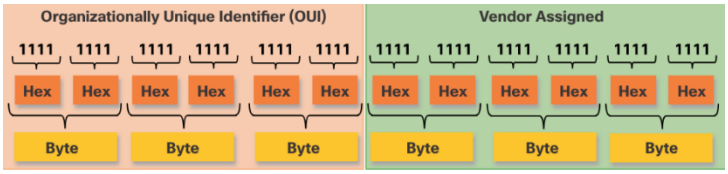
Minimum frame size: 64 bytes
Maximum frame size: 1518 bytes
في الحساب Preamble **NOT included**

Field	Size
Preamble + SFD	8 bytes
Destination MAC	6 bytes
Source MAC	6 bytes
Type / Length	2 bytes
Data	46–1500 bytes
FCS	4 bytes



Important :Frame < 64 bytes = Runt or collision fragment it's (dropped) , Frame > 1500 bytes data = Jumbo frame or "baby giant frames" , Jumbo frames are usually supported by most Fast Ethernet and Gigabit Ethernet switches and NICs .

MAC Address Basics :
48 bits = 6 bytes
12 hexadecimal digits مكتوبة بـ
مثال : 00-1A-2B-3C-4D-5E



OUI (Organizationally Unique Identifier)
- **24 bits (3 bytes)** أول
- تحدد الشركة المصنعة
- **IEEE** مسجلة لدى

[VERY IMPORTANT] Every MAC address must be globally unique

Frame Processing :
عند استقبال Frame :
1 فحص Destination MAC
2 إذا **لا يطابق** Drop
3 إذا **يطابق** De-encapsulation
4 Broadcast / Multicast → Accept

MAC Address Types :

1- Unicast
- One-to-One
- **Unicast** دائمًا Source MAC
- يربط ARP (IPv4) / ND (IPv6) IP بـMAC

2- Broadcast
- One-to-All
- Destination MAC: FF-FF-FF-FF-FF-FF
- يصل لكل الأجهزة داخل الـLAN
- لا يمر عبر Router

Broadcast يوصّل لكل الأجهزة في LAN ما يطلع خارج الشبكة

3- Multicast
- One-to-Many (Group)
- IPv4: 01-00-5E-xx-xx-xx
- IPv6: 33-33-xx-xx-xx-xx
- يُرسل لمجموعة محددة
- Unicast دائمًا Source MAC

[EXAM TRAP] Multicast & Broadcast are flooded by switches

Switch Fundamentals
- **Layer 2** يعمل في Switch
- لا يهتم بنوع البيانات(IP / ARP / IPv6)

- يعتمد فقط على :
  - **Source MAC**
  - **Destination MAC**
- جدول MAC يسمى أيضًا :
  - **CAM Table**

عند دخول Frame:

1 يفحص **Source MAC**

2 يضيفها للجدول مع Port number

3 يعطي timer افتراضي 5- minutes

[EXAM FAVORITE] Switch learns ONLY from Source MAC

Forwarding Decisions :

- Forward out specific port : **معروف** Destination MAC
- Flood (Unknown Unicast) : **غير معروف** Destination MAC
- Broadcast / Multicast: Flood

Filtering Frames :

عند معرفة Destination MAC :

- يمنع الإرسال لبقية المنافذ
- يرسل فقط للمنفذ الصحيح

Frame Forwarding Methods :

1 **Store-and-Forward**

- يستقبل Frame كامل
- يفحص **CRC**
- يرسل فقط الإطارات السليمة
- مطلوب لـ:
  - **QoS**
  - VoIP

2 **Cut-Through**

- يرسل قبل اكتمال Frame
- لا يفحص الأخطاء

There are two variants of cut-through switching :

- **Fast-Forward (Latency أقل)**
- **Fragment-Free (يفحص أول ٦٤ bytes)**

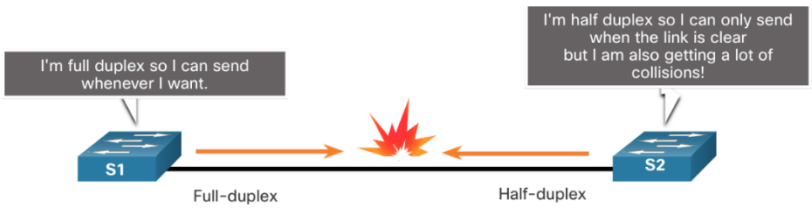
Important : Store-and-Forward = reliable , Cut-Through = low latency

Memory Buffering :

Type	Description
Port-based	Queue لكل Port
Shared memory	Memory مشتركة (أفضل)

Duplex & Speed :

- **Full-Duplex**: Send & Receive
- **Half-Duplex**: One direction only
- Gigabit Ethernet: Full-Duplex only



Duplex Mismatch : (سؤال شائع)

سبب شائع لمشاكل الأداء

يحدث عند : Full طرف موصل بطرف Half

[EXAM TIP] Best practice: both sides auto-negotiation ON

Auto-MDIX :

- يحدد تلقائيًا نوع الكيبل
- Enabled by default (IOS ≥ 12.2(18)SE)
- لا تعتمد عليه ١٠٠%

Address Resolution L2

كل جهاز في شبكة Ethernet يمتلك:

Layer	Address	Purpose
Layer 2	**MAC Address**	NIC → NIC (Local)
Layer 3	**IP Address**	End-to-End Delivery

[EXAM FAVORITE] MAC = Local delivery , IP = End-to-End delivery

Destination on Same Network :

إذا كان **Destination IP داخل نفس الشبكة:**

- Destination MAC = **MAC الخاص بالجهاز الهدف**
- لا يتم استخدام الـRouter

Destination on Remote Network :

إذا كان **Destination IP في شبكة مختلفة:**

- Destination MAC = **MAC الخاص بالـDefault Gateway**
- IP Destination يبقى **IP الجهاز النهائي**

+ Local → MAC = target device

+ Remote → MAC = default gateway

Address Resolution Protocols :

- **IPv4** يستخدم **ARP** :
- **IPv6** يستخدم **Neighbor Discovery (ICMPv6)** :

[EXAM FAVORITE] ARP = IPv4 , ND = IPv6

ARP Purpose :

يُستخدم ARP من أجل:

- تحويل **IPv4 Address → MAC Address**
- الاحتفاظ بالنتائج داخل:
  - **ARP Table / ARP Cache**

ARP Functions :

عند إرسال Packet :

1. البحث في **ARP Table**
2. إذا وُجد → MAC يُستخدم مباشرة
3. إذا لم يوجد:
   - إرسال **ARP Request (Broadcast)**
4. الجهاز المطلوب يرد بـ:
   - **ARP Reply (Unicast)**
5. يتم حفظ النتيجة في ARP Table

[EXAM FAVORITE] ARP Request = Broadcast , ARP Reply = Unicast

ARP في Local vs Remote :

- **Local Network** :
  - ARP لـDestination IP
- **Remote Network** :
  - ARP لـ **Default Gateway IP**

ARP Frame Details :

- Destination MAC في ARP Request = FF:FF:FF:FF:FF:FF

ARP Cache / Table :

- تحتوي :
  - IPv4 ↔ MAC mappings
- غير دائمة

ARP Timer :

- تنتهي بعد مدة (مثال: ~١٥–٢٠ دقيقة)
- تختلف حسب نظام التشغيل

ARP Issues :

- ARP Requests :
  - تصل لكل الأجهزة في الشبكة

- كثرتها :
  - تقلل الأداء
- **ARP Spoofing / Poisoning :**
  - هجوم أمني
- Enterprise Switches:
  - توفر حماية DAI

[EXAM POINT] ARP has NO authentication

What is ND?

بروتوكول **IPv6 Neighbor Discovery** يوفر:

- Address Resolution
- Router Discovery
- Redirect services

ND vs ARP :

Feature	ARP	ND
IP Version	IPv4	IPv6
Protocol	ARP	ICMPv6
Broadcast	Yes	No
Multicast	No	Yes

[EXAM FAVORITE] IPv6 does NOT use broadcast

ND Messages :

Message	Purpose
NS	Neighbor Solicitation
NA	Neighbor Advertisement
RS	Router Solicitation
RA	Router Advertisement
Redirect	Better next-hop

IPv6 Address Resolution (ND Process) :

1 IPv6 يعرف Host

2 يرسل **NS (Multicast)**

3 Target يرد بـ (Unicast) NA

4 يتم حفظ MAC في Neighbor Table

Network Layer L3

The network layer performs four basic operations :

- Addressing end devices
- Encapsulation
- Routing
- De-encapsulation

[EXAM FAVORITE] Layer 3 protocols = Routed Protocols

IP encapsulates the transport layer segment.

IP can use either an IPv4 or IPv6 packet and not impact the layer 4 segment.

The IP addressing does not change from source to destination.

IP يوصف بأنه:

Characteristic	Meaning
Connectionless	لا ينشئ اتصال
Best Effort	لا يضمن التسليم
Media Independent	يعمل على أي وسط

IP is Connectionless :

- Handshake لا يوجد
- Acknowledgment لا يوجد
- Synchronization لا يوجد

إذا احتجنا اتصال موثوق **TCP** يتكفل بذلك

[COMMON TRAP] IP is NOT reliable

IP is Best Effort :

- لا يضمن وصول الحزم
- لا يعيد الإرسال
- لا يعرف إن كانت الحزمة وصلت
- لا ينتظر ACK

IP is unreliable:

- It cannot manage or fix undelivered or corrupt packets.
- IP cannot retransmit after an error.
- IP cannot realign out of sequence packets.
- IP must rely on other protocols for these functions.

Media أي عبر IP Packet يمكن إرسال : Media Independent

: يحدد Network Layer •

- **MTU**

- MTU يُبلغ من Data Link Layer

Fragmentation :

- **IPv4** يحدث في

يؤدي إلى Latency

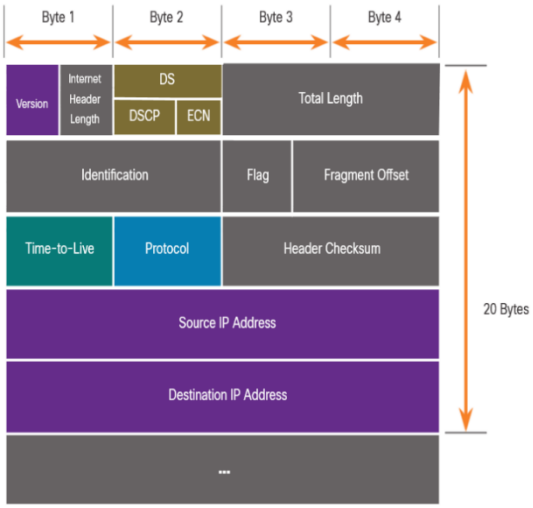
- **IPv6 لا يدعم fragmentation في الراوتر**

[EXAM FAVORITE] IPv6 routers do NOT fragment packets

IPv4 Header :

- Binary مكتوب
- Layer 3 يُفحص من جميع أجهزة

اهم حقلين الـ source IP والـ destination IP



IPv4 Header Fields :

Field	Function
Version	IPv4 = 0100
Differentiated Services / Type of Service	Used for QoS
Header Checksum	كشف أخطاء الهيدر
TTL	Hop count
Protocol	TCP / UDP / ICMP
Source IP	عنوان المصدر
Destination IP	عنوان الوجهة

[EXAM FAVORITE] TTL decreases by 1 at every router

مشاكل IPv4 :

- address depletion : نفاد العناوين
- Lack of end-to-end connectivity : الاعتماد على NAT
- Increased network complexity – NAT : تعقيد الشبكة

IPv6 Overview :

- **IETF** طُور بواسطة
- **128 bits**

مزايا :

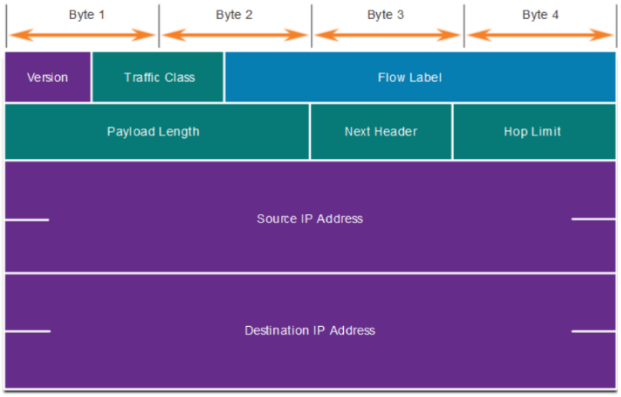
- مساحة عناوين ضخمة
- Header مبسط
- لا حاجة لـ NAT

IPv6 Header :

- **40 bytes** ثابت الطول
- أبسط من IPv4
- لا يحتوي :
  - Header Checksum
  - Fragment Offset
  - Flags

IPv6 Header Fields أهم :

Field	Function
Version	IPv6 = 0110
Traffic Class	QoS
Flow Label	معالجة تدفق معين
Payload Length	طول البيانات
Next Header	TCP / UDP / ICMPv6
Hop Limit	بديل TTL
Source IP	128-bit
Destination IP	128-bit



IPv6 packet may also contain extension headers (EH).
EH headers characteristics:
• provide optional network layer information
• are optional • are placed between IPv6 header and the payload
• may be used for fragmentation, security, mobility support, etc.
[VERY IMPORTANT] IPv6 fragmentation is done ONLY by the source

Host Forwarding Decision :

Routing Table كل Host يملك
الوجهات الممكنة :

- **Itself**
  - IPv4: 127.0.0.1
  - IPv6: ::1
- **Local Host**
  - Same LAN
- **Remote Host**
  - Different network

Default Gateway (DGW) :

- Layer 3 Switch أو Router
- يجب أن يكون :
  - بنفس شبكة الـLAN
- بدون DGW لا يمكن الخروج خارج الشبكة

Router Forwarding Decision :

عند استقبال Frame :

1. إزالة Layer 2 Header
2. فحص Destination IP
3. البحث في Routing Table
4. إعادة Encapsulation
5. إرسال للـNext Hop

Router Routing Table Types :

Route Type	Description
Directly Connected	تلقائي
Remote	Static أو Dynamic

Route Type	Description
Default Route	عند عدم التطابق

Static Routing :

- configured manually : يُضبط يدويًا
- adjusted manually : لا يتكيف مع التغييرات
- Good for small non-redundant networks : مناسب للشبكات الصغيرة
- Default Route غالبًا

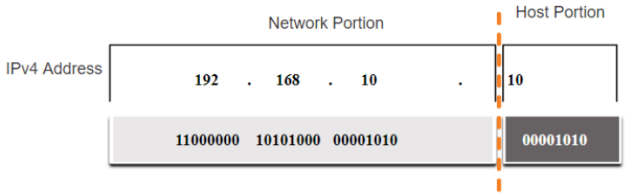
Dynamic Routing :

- Discover remote networks : يكتشف الشبكات تلقائيًا
- Choose the best path : يختار أفضل مسار
- Maintain up-to-date information : يتكيف مع التغييرات

IPv4 L3

IPv4 Address Basics :

- **32 bits** عنوان IPv4 طوله
  يتكون من :
  - **Network Portion**
  - **Host Portion**
  التقسيم يتم باستخدام :
  - **Subnet Mask**



[EXAM FAVORITE] IPv4 address = Network part + Host part

The subnet mask is compared to the IPv4 address bit for bit, from left to right.

Subnet Mask :

IP **bit by bit** يُقارن مع عنوان

Logical AND العملية تسمى

1 AND 1 = 1 anything else = 0

Network Address استخراج : الهدف

A prefix length is a less cumbersome method used to identify a subnet mask address.

It is written in “slash notation” therefore, count the number of bits in the subnet mask and prepend it with a slash.

في كل شبكة يوجد :

Type	Description
Network Address	0 = bits host كل
First Host	Network + 1
Last Host	Broadcast - 1
Broadcast Address	1 = bits host كل

		Network Portion			Host Portion	Host Bits
IPv4 addr	Subnet mask 255.255.255.0 or /24	255 11111111	255 11111111	255 11111111	0 00000000	
Subn	Network address 192.168.10.0 or /24	192 11000000	168 10100000	10 00001010	0 00000000	All 0s
	First address 192.168.10.1 or /24	192 11000000	168 10100000	10 00001010	1 00000001	All 0s and a 1
IPv4 netw addr	Last address 192.168.10.254 or /24	192 11000000	168 10100000	10 00001010	254 11111110	All 1s and a 0
	Broadcast address 192.168.10.255 or /24	192 11000000	168 10100000	10 00001010	255 11111111	All 1s

[EXAM TRAP] Network & Broadcast addresses are NOT usable

IPv4 Unicast, Broadcast, Multicast :

Unicast :

- إرسال إلى **جهاز واحد فقط**
- مثال : PC → Printer

Broadcast :

- إرسال إلى **كل الأجهزة في نفس الشبكة**
- مثال :
  - DHCP Discover
  - ARP Request
- عنوان:Broadcast
  - **آخر عنوان في الشبكة**

[EXAM FAVORITE] Routers do NOT forward broadcasts

Multicast :

إرسال إلى **مجموعة محددة**

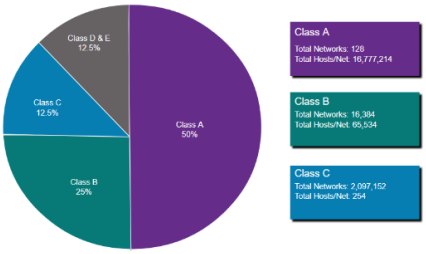
Multicast IPv4 نطاق
224.0.0.0 – 239.255.255.255
Public vs Private IPv4 :
Public :
- قابلة للتوجيه عبر الإنترنت
- فريدة عالميًا
- تُستخدم معISPs

نطاقات Private IPv4 من RFC 1918 :
10.0.0.0/8
172.16.0.0 – 172.31.255.255 (/12)
192.168.0.0/16
[EXAM FAVORITE] Private IPs are NOT routable on the internet
NAT (Network Address Translation)
- translates the internal private address to a public global IP address : Private → Public يحول
- **Edge Router** يعمل عادة على

Special IPv4 Addresses :
1- Loopback
127.0.0.0/8
Common: 127.0.0.1
- لاختبار TCP/IP محليًا

2- Link-Local (APIPA) Automatic Private IP Addressing
169.254.0.0/16
- يُستخدم عند فشل DHCP

Legacy Classful Addressing :
- Class A (0.0.0.0/8 to 127.0.0.0/8)
- Class B (128.0.0.0 /16 – 191.255.0.0 /16)
- Class C (192.0.0.0 /24 – 223.255.255.0 /24)
- Class D (224.0.0.0 to 239.0.0.0)
- Class E (240.0.0.0 – 255.0.0.0)



Classful addressing wasted many IPv4 addresses.
Class D for Multicast and Class E for Experimental.
Classful address allocation was replaced with classless addressing which ignores the rules of classes (A, B, C).
[IMPORTANT] Modern networks use Classless addressing (CIDR)
IP Address Allocation :
- **IANA** توزع العناوين على :
  - **RIRs**
  - RIRs توزعها على :
    - ISPs
    - Organizations

Broadcast Domains :
Broadcast يمرر : Switch
Broadcast يوقف : Router
[EXAM FAVORITE] Router = The only device that stops broadcasts, do not propagate broadcasts.
Problems with Large Networks :
- Broadcast كثرة
- انخفاض الأداء

Solution :
Subnetting : تقسيم الشبكة إلى Broadcast Domains أصغر
Reasons for Subnetting :
- تحسين الأداء

- تقليل Broadcast
- تطبيق سياسات أمنية
- تقسيم حسب :
  - الموقع
  - الوظيفة
  - نوع الجهاز

Reasons for Segmentation

- تحسين الأداء
- تقليلBroadcast
- تطبيق سياسات أمنية
- تقسيم حسب:
  - الموقع
  - الوظيفة
  - نوع الجهاز

Subnetting on Octet Boundary :

Prefix	Hosts
/8	16,777,214
/16	65,534
/24	254

Subnetting within Octet (/24) :

Prefix	Subnets	Hosts
/25	2	126
/26	4	62
/27	8	30
/28	16	14
/29	32	6
/30	64	2

[EXAM FAVORITE] /30 is used for point-to-point links

[EXAM FAVORITE] You can NOT borrow all host bits [can't use 11111111 = \32]

Efficient Subnetting :

- الهدف :
  - تقليل العناوين المهدرة
  - تلبية عدد hosts المطلوب

[EXAM FAVORITE] VLSM = subnet a subnet

Within a network, there are different types of devices that require addresses:

• End user clients – Most use DHCP to reduce errors and burden on network support staff. IPv6 clients can obtain address information using DHCPv6 or SLAAC.

• Servers and peripherals – These should have a predictable static IP address.

• Servers that are accessible from the internet – Servers must have a public IPv4 address, most often accessed using NAT.

• Intermediary devices – Devices are assigned addresses for network management, monitoring, and security.

• Gateway – Routers and firewall devices are gateway for the hosts in that network.

IPv6 L3

لماذا IPv6 ؟

- **IPv4 addresses are running out**
- IPv6 هو **الخليفة الرسمي** لـIPv4
- IPv6 يستخدم :
  - **128-bit address space**
- IPv6 حل مشاكل :

- Limited IPv4 space
- NAT complexity
- IoT expansion

[EXAM FAVORITE] IPv6 was created because IPv4 address space is exhausted

IPv4 & IPv6 Coexistence :

الانتقال لـ IPv6 سيأخذ وقت، لذلك يوجد ٣ طرق :

Method	Description
Dual Stack	تشغيل IPv4 و IPv6 معًا
Tunneling	IPv4 داخل IPv6
Translation	NAT64 (IPv6 ↔ IPv4)

الهدف النهائي من IPv6 هو: تشغيل IPv6 بشكل مباشر (Native) من المصدر إلى الوجهة مو الاعتماد الدائم على حلول مؤقتة مثل Tunneling أو Translation

[EXAM POINT] Goal = Native IPv6 (NOT tunneling forever)

IPv6 Basics :

- الطول **128 bits**
- مكتوب بالـ **Hexadecimal**
- غير حساس لحالة الأحرف
- كل جزء **Hextet (16 bits)**
- الشكل الأساسي x:x:x:x:x:x:x:x

Rule 1 – Omit Leading Zeros :

حذف الأصفار **في بداية** الـ hextet فقط لا تحذف الأصفار في النهاية

مثال: 0db8 → db8

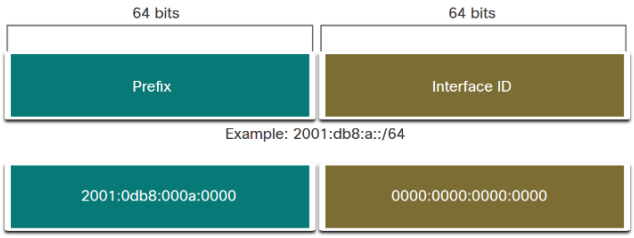
0000 → 0

Type	Format
Preferred	2001 : 0db8 : 0000 : 1111 : 0000 : 0000 : 0000 : 0200
Compressed	2001:db8:0:1111::200

Rule 2 – Double Colon (::)

- تُستخدم لتعويض:
  - **مجموعة متتالية من hextets كلها 0**
- تُستخدم **مرة واحدة فقط**

[EXAM TRAP] :: can be used ONLY ONCE



IPv6 Categories :

- **Unicast**
- **Multicast**
- **Anycast**

No Broadcast in IPv6.

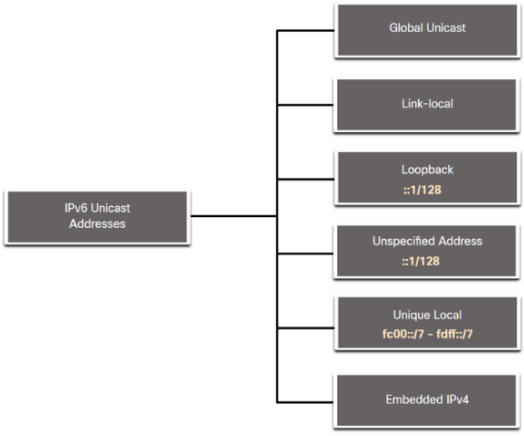
في التقسيم مثل ipv4 ولاكن تختلف المسميات الprefix هي الnetwork portion وهكذا

IPv6 Prefix Length :

- يستخدم Slash notation
- من **0/** إلى **128/**
- الموصى به للشبكات 64/

Anycast – This is any IPv6 unicast address that can be assigned to multiple devices •
A packet sent to an anycast address is routed to the nearest device having that Address.

IPv6 Unicast Types :



1- Global Unicast Address (GUA)

- مثل Public IPv4
- Routable on Internet
- النطاق الحالي 2000::/3

2- Link-Local Address (LLA)

- إجباري لكل interface
- للتواصل داخل نفس الشبكة فقط
- غير قابل للتوجيه
- النطاق fe80::/10

تخيل:

- **LLA باب البيت**
- **GUA عنوان الشخص اللي بتزوره**

عشان تطلع :

- تفتح باب بيتك (LLA)
- ثم تسافر لعنوان الشخص (GUA)

LLA لا يُستخدم كـ Destination IP للإنترنت و LLA يُستخدم فقط كـ(Next-Hop (DG default getway

[EXAM FAVORITE] Default gateway in IPv6 = Router LLA

3- Unique Local Address (ULA)

- مشابه Private IPv4 لكن :
  - لا يُستخدم للأمان
- النطاق fc00::/7 – fdff::/7

IPv6 GUA Structure :

يتكون من :

1 Global Routing Prefix

2 Subnet ID

3 Interface ID (64 bits)

Dynamic Addressing for IPv6 GUAs :

RS & RA Messages :

- **RS** () من Host → Router
- **RA**: من Router → Host

RS (Router Solicitation) :

من Host **إلى** Router

المعنى *يا راوتر، علّمني كيف أضبط IPv6*

RA (Router Advertisement) :

من Router **إلى** Host

الراوتر يعلن فيه :

- **Prefix** شبكة IPv6
- **Prefix Length** غالبًا /64
- **Default Gateway (LLA للراوتر)**
- **DNS info (أحيانًا)**

IPv6 GUA Assignment Methods :

1 **SLAAC**

- بدون DHCPv6
- يستخدم :
  - Prefix من RA
  - Interface ID (EUI-64 أو Random)

2 **SLAAC + Stateless DHCPv6**

- SLAACللعنوان
- DHCPv6 للمعلومات(DNS)

3 **Stateful DHCPv6**

- DHCPv6يعطي:
  - Address
  - DNS
  - Domain
- Router LLA = Default Gateway

[EXAM FAVORITE] RA decides the addressing method

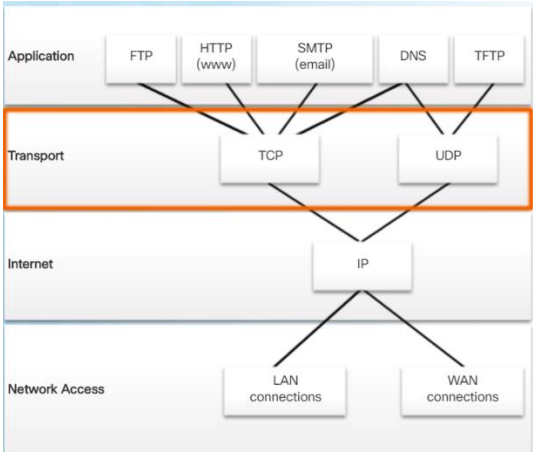
Transport Layer L4

طبقة النقل (Layer 4) مسؤولة عن :

- **Logical communication between applications**
- الربط بين:
  - **Application Layer**
  - والطبقات السفلى(Network & below)

[EXAM FAVORITE] Transport Layer = Application-to-Application communication

Transport Layer Responsibilities :



- Tracking individual conversations
- **Segmentation** وإعادة التجميع
- **Transport Header** إضافة
- **Multiplexing (تشغيل عدة تطبيقات في نفس الوقت)**
- إدارة المحادثات المتعددة على نفس الشبكة

Important: Multiplexing uses port numbers

Transport Layer Protocols :

- IP لا يحدد كيف يتم النقل
- Transport Layer تحدد :
  - كيف تُنقل البيانات
  - مدى الموثوقية
- البروتوكولات :
  - **TCP**
  - **UDP**

TCP Characteristics :

TCP هو:

- **Connection-oriented**
- **Reliable**
- **Stateful**

TCP Basic Operations :

- Numbering segments
- Tracking sent data
- **Acknowledgment (ACK)**
- Retransmission
- Sequencing
- Flow control

[EXAM FAVORITE] TCP provides Reliability + Sequencing + Flow Control

TCP Features :

- Establishing a session
- Reliable delivery
- Same-order delivery
- Flow control

TCP Control Flags :

Flag	Function
SYN	Start session
ACK	Acknowledge
FIN	End session
RST	Reset
PSH	Push data
URG	Urgent data

[EXAM TRAP] SYN & FIN are never used together

Applications that use TCP :

- HTTP / HTTPS
- FTP
- SMTP
- SSH
- IMAP / POP3

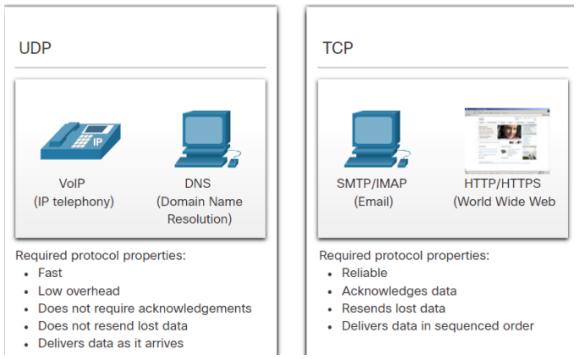
UDP Characteristics :

: هو UDP

- **Connectionless**
- **Best effort**
- **Low overhead**
- No reliability
- No sequencing
- No retransmission

+ UDP = Fast

- UDP = Not reliable



Applications that use UDP :

- VoIP
- Live streaming
- DNS
- DHCP
- SNMP
- TFTP

[EXAM FAVORITE] Real-time apps prefer UDP

Why Port Numbers ? :

- **multiple simultaneous conversations** إدارة
- التمييز بين التطبيقات

Source vs Destination Port :

- Source Port:
  - يحدد التطبيق المحلي
  - **Ephemeral** غالبًا
- Destination Port:
  - يحدد التطبيق البعيد
  - **Well-known** غالبًا

Well-Known Port Numbers

Port Number	Protocol	Application
20	TCP	File Transfer Protocol (FTP) - Data
21	TCP	File Transfer Protocol (FTP) - Control
22	TCP	Secure Shell (SSH)
23	TCP	Telnet
25	TCP	Simple Mail Transfer Protocol (SMTP)
53	UDP, TCP	Domain Name Service (DNS)
67	UDP	Dynamic Host Configuration Protocol (DHCP) - Server
68	UDP	Dynamic Host Configuration Protocol - Client
69	UDP	Trivial File Transfer Protocol (TFTP)
80	TCP	Hypertext Transfer Protocol (HTTP)
110	TCP	Post Office Protocol version 3 (POP3)
143	TCP	Internet Message Access Protocol (IMAP)
161	UDP	Simple Network Management Protocol (SNMP)
443	TCP	Hypertext Transfer Protocol Secure (HTTPS)

Socket :

Socket = IP Address + Port Number

: يسمح

- Multiple clients
- Multiple sessions

Port Number Ranges :

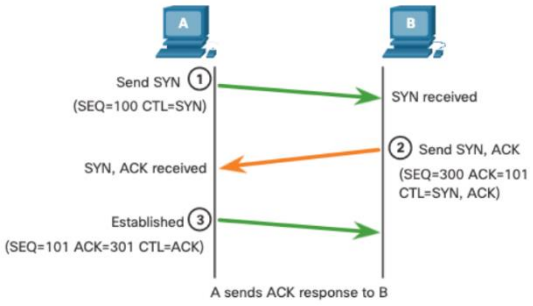
Range	Type
0 – 1023	Well-known
1024 – 49151	Registered
49152 – 65535	Dynamic or private

TCP Communication Process :

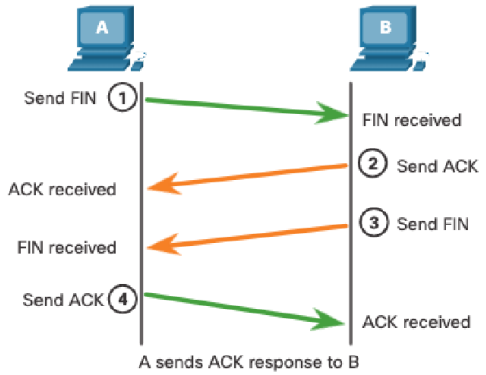
TCP Server Processes :

- كل خدمة تستخدم **Port واحد فقط**
- لا يمكن خدمتين على نفس الـPort
- Listening = مفتوح Port

TCP Connection Establishment (3-Way Handshake) :



1 SYN
2 SYN-ACK
3 ACK
[EXAM FAVORITE] TCP uses Three-Way Handshake



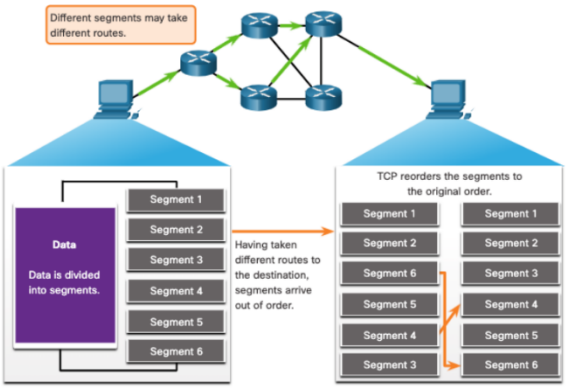
TCP Session Termination :

1 FIN
2 ACK
3 FIN
4 ACK

Reliability and Flow Control :

1- TCP Reliability :

Uses :
- Sequence Numbers
- ACKs
- يعيد ترتيب البيانات
- يعيد الإرسال عند الفقد



2- Retransmission & SACK :

- **SACK** = Selective Acknowledgment
- يحدد القطع المفقودة فقط
- يتم التفاوض عليه أثناء الـHandshake

3- Flow Control :

- يعتمد على :
  - **Window Size**
- يمنع إرهاق المستقبل

4- MSS (Guaranteed Question) :

- MSS = Maximum Segment Size
- Ethernet MTU = 1500 bytes
- IPv4 Header = 20 bytes
- TCP Header = 20 bytes
- **ليش MSS مهم ؟**
- **Fragmentation** يمنع
- يخلي كل TCP Segment يركب داخل Frame واحد بدون تقطيع

MSS is 1,460 bytes using 1500-40.

5- Congestion Control :

- عند ازدحام الشبكة :
  - يقلل سرعة الإرسالTCP
- يستخدم :
  - Timers
  - Algorithms

UDP Datagram Reassembly :

- لا يستخدم sequence numbers
- يعيد البيانات حسب وصولها
- لا يعيد الترتيب

UDP Client / Server :

- Server:
  - Well-known or registered port
- Client:
  - Dynamic (ephemeral) port

Application Layer L5-6-7

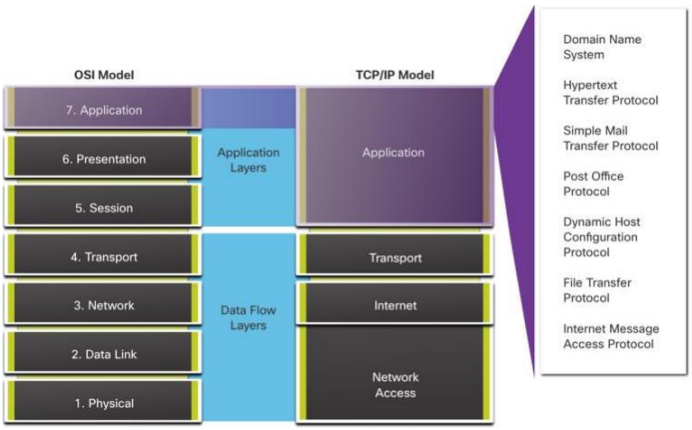
- الطبقات الثلاث العليا في OSI :

- **Application**
- **Presentation**
- **Session**
- تقابلها طبقة واحدة في TCP/IP :
- **Application Layer**

[EXAM FAVORITE] OSI Layers 7–6–5 = TCP/IP Application Layer

Application Layer توفر **واجهة بين التطبيقات والشبكة**

- التطبيقات لا تتعامل مع الشبكة مباشرة
- البروتوكولات الأشهر:
  - HTTP
  - FTP
  - TFTP
  - IMAP
  - DNS



+ Application Layer provides services
- Applications themselves are NOT protocols

Presentation Layer (Layer 6) :

ثلاث وظائف **مضمونة بالاختبار**:

1. Formatting (تنسيق البيانات)
2. Compression (ضغط البيانات)
3. Encryption / Decryption (تشفير وفك تشفير)

[EXAM FAVORITE] Encryption = Presentation Layer

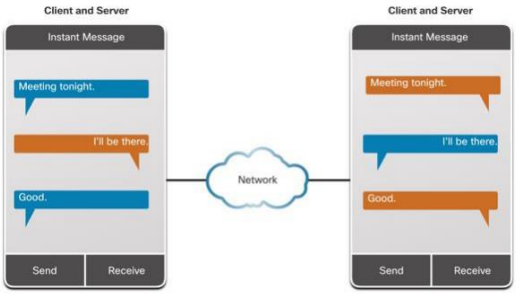
Session Layer (Layer 5) :

- إنشاء الجلسات
- الحفاظ عليها
- إعادة تشغيل الجلسات المتوقفة
- إدارة الـDialogs

[COMMON TRAP] Session Layer ≠ Transport Layer

Peer-to-Peer (P2P) Network :

- لا يوجد Server مخصص
- كل جهاز:
  - Client + Server بنفس الوقت
- الأدوار Roles تتغير حسب الطلب



[EXAM FAVORITE] In P2P, roles are per-request

P2P Applications :

- بعض الأنظمة :
  - Hybrid (فقط Index Server)
- أمثلة مشهورة :
  - BitTorrent
  - eDonkey
  - Freenet
  - Direct Connect

HTTP & HTML (Web) :

- عند كتابة URL :

1. المتصفح يحدد :
   - Protocol
   - Server name
   - File name
2. DNS يحول الاسم إلى IP
3. HTTP GET request

4 السيرفر يرسل HTML
5 المتصفح يعرض الصفحة

HTTP Basics:

- Protocol :
  - **Request / Response**
- أشهر الرسائل :
  - **طلب بيانات → GET**
  - **إرسال بيانات → POST**
  - **رفع موارد → PUT**

[EXAM FAVORITE] HTTP is NOT secure , HTTPS = HTTP + Encryption

Email Protocols :

1- SMTP

- إرسال البريد
- Port: TCP 25
- Store-and-forward
- يستخدم بين :
  - Client → Server
  - Server → Server

2- POP3

- استلام البريد
- Port: TCP 110
- **يحذف الرسائل من السيرفر**
- غير مناسب للنسخ الاحتياطي

3- IMAP

- استلام البريد
- **الرسائل تبقى على السيرفر**
- مزامنة بين الأجهزة
- أفضل للشركات

Important: POP = download & delete , IMAP = sync & keep

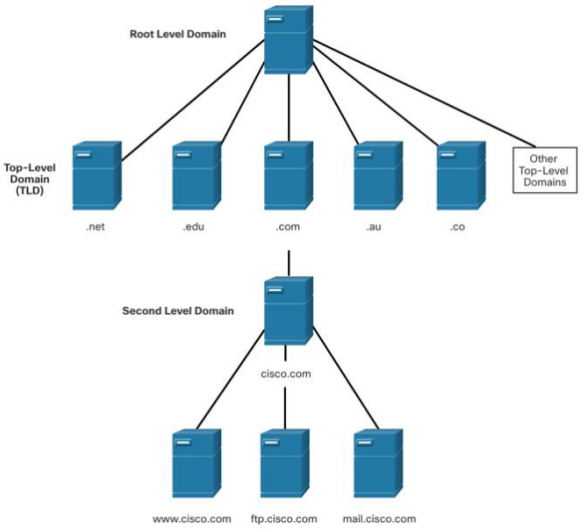
IP Addressing Services :

DNS (Domain Name System)

- يحول:
  - Name → IP
- يستخدم :
  - FQDN (Fully-qualified domain names)
- أسهل للبشر

DNS Record Types :

Record	Function
A	IPv4 address
AAAA	IPv6 address
NS	Name Server
MX	Mail Server



DNS Hierarchy :

- Root
- Top-Level Domains (.com, .org, .sa)
- Second-Level Domains
- Subdomains

[EXAM FAVORITE] DNS is hierarchical, not flat

Examples of top-level domains:

• com - a business or industry

- org - a non-profit organization
- au - Australia

nslookup Command :

- اختبار DNS
- استكشاف الأخطاء

DHCP (IPv4) :

- توزيع تلقائي لـ:
  - IP
  - Subnet Mask
  - Gateway
  - DNS

DHCP vs Static :

- DHCP:
  - End devices
- Static:
  - Routers
  - Servers
  - Printers

DHCP Process (DORA) :

Discover
Offer
Request
Acknowledge

[EXAM FAVORITE] DHCP uses broadcast at first

DHCPv6 :

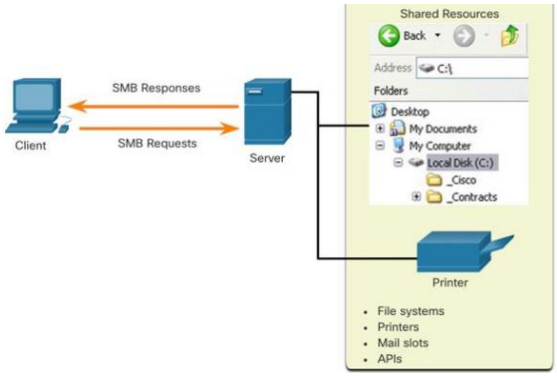
- يشبه DHCPv4
- لا يعطي Default Gateway
- Default Gateway يأتي من :
  - **Router Advertisement**

File Sharing Services :

FTP

- نقل ملفات
- يستخدم اتصالين :
  - Control → TCP 21
  - Data → TCP 20

[EXAM FAVORITE] FTP uses TWO connections

SMB (Server Message Block)

- File & Printer sharing
- Client/Server
- اتصال طويل
- المستخدم يتعامل مع الملفات كأنها محلية